

A well-posed gap in the two-color lattice literature: $T_c/\sqrt{\sigma}$ for SU(2) with $N_f = 3$ light fundamental flavours

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Two-color QCD has a substantial finite-temperature literature at $N_f = 2$, yet no published determination of the transition temperature in string-tension units, $T_c/\sqrt{\sigma}$, exists for $N_f = 3$ light fundamental Dirac flavours. We state the computation cleanly: separate chiral and deconfinement crossovers, a chiral extrapolation, a continuum statement at least two temporal extents, and an explicit confrontation of the staggered-rooting versus Wilson trade-off in a pseudo-real theory. The gap is scientifically interesting independent of any particular dark-sector model: flavour dependence of $T_c/\sqrt{\sigma}$ is well mapped in SU(3), while SU(2)'s enlarged chiral pattern and different Goldstone count leave the saturation of that flavour dependence open. We record, for transparency only, a publicly pre-registered numerical stake on the chiral transition and the precision required to score it; the stake is not the result of this note. The result of the note is that the calculation is conventional, well posed, and missing.

I. THE MISSING NUMBER

We ask for a finite-temperature lattice determination of

$$T_c/\sqrt{\sigma} \quad (1)$$

in SU(2) gauge theory with $N_f = 3$ degenerate light Dirac fermions in the fundamental representation, with four requirements stated up front.

1. *Both transitions separately.* The chiral crossover (peak of the chiral susceptibility) and the deconfinement crossover (peak of the Polyakov-loop susceptibility). In two-color QCD the Polyakov peak weakens as the quarks lighten, opposite to the three-color behaviour [1], so the splitting is itself a result.
2. *A chiral extrapolation.* Ensembles at $m_{\text{PS}}/\sqrt{\sigma} \lesssim 1$ with the quark-mass slope measured. The SU(3) benchmark is $T_c/\sqrt{\sigma} = 0.40(1) + 0.039(4)(m_{\text{PS}}/\sqrt{\sigma})$ [2]; whether a pseudo-real theory follows the same pattern is untested.
3. *A scale set by the string tension* directly, or by $r_0/t_0/w_0$ with the conversion stated on the same ensembles.
4. *A continuum statement* from at least two temporal extents ($N_\tau = 6, 8; 10\text{--}12$ if affordable). Coarse-lattice SU(3) values drifted downward by about 10% in the continuum; whether SU(2) does the same is unknown.

II. WHY THE GAP IS INTERESTING WITHOUT ANY MODEL

The literature at $N_f = 2$ is not empty. On the staggered side, Astrakhantsev *et al.* [3] report $\sqrt{\sigma_0} = 476(5)$ MeV on-ensemble, and Kudrov, Bornyakov and Goy [4] place the deconfinement crossover at $T_d(0) = 230(10)$ MeV, equivalently $r_0 T_d(0) = 0.55(4)$. Taken together these give $T_d/\sqrt{\sigma} \approx 0.48$. On the Wilson side, two determinations of the chiral crossover disagree at the $\sim 30\%$ level [5, 6].

What is missing is $N_f = 3$ light fundamentals. $N_f = 4$ exists only in qualitative form from older work. That absence matters for three reasons that do not require a dark-sector story.

Flavour dependence in a pseudo-real theory. In SU(3) the flavour suppression of $T_c/\sqrt{\sigma}$ largely saturates after the first two flavours. SU(2)'s fundamental representation is pseudo-real: the chiral symmetry is enlarged, $\text{SU}(2N_f) \rightarrow \text{Sp}(2N_f)$, the Goldstone count differs (fourteen at $N_f = 3$), and whether the SU(3) saturation pattern transfers is an open, clean question.

Conformal-window safety. $N_f = 3$ sits well below the SU(2) fundamental window edge near $N_f \approx 6$ [7, 8], so the theory confines and breaks chiral symmetry. The computation is conventional.

Cost. Two colors, three light flavours, moderate volumes; existing $N_f = 2$ ensembles and tuning are a natural starting point.

III. THE DISCRETIZATION IS THE HARD CHOICE

$N_f = 3$ is the least comfortable flavour count to discretize, and anyone costing the computation will see that first.

Staggered. Three flavours require the odd root, $\det^{3/4}$ — rooted staggered where it is least comfortable. Two-

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color QCD adds a second problem: with fundamental quarks the staggered action does not reproduce the continuum symmetry-breaking pattern [3]. Since the pseudo-real enlargement is precisely what makes the N_f dependence interesting, a discretization that deforms it is not a neutral choice.

Wilson. The symmetry pattern is correct, at the cost of explicit chiral-symmetry breaking at finite lattice spacing — which is exactly what a chiral-crossover determination is trying to measure — and the two existing $N_f = 2$ Wilson determinations already disagree at $\sim 30\%$.

Neither choice is settled here. The discretization is part of the computation, not a detail beneath it. A result whose discretization is not argued for will not settle the flavour question at the precision discussed next.

IV. A PRE-REGISTERED STAKE (TRANSPARENCY ONLY)

An independent phenomenological program has publicly pre-registered a numerical stake on the chiral transition, placed before any dedicated $N_f = 3$ computation existed: $T_c/\sqrt{\sigma} = 0.34657$ ($= \frac{1}{2} \ln 2$), with pre-committed kill conditions. The stake is recorded here for transparency, not as a result of this note.

The precision required to score the stake is more im-

portant than the central value. The stake distinguishes two hypotheses 0.44% apart; a determination with relative uncertainty $\sigma > 0.44\%$ cannot score either way and is registered in advance as inconclusive. Discrimination needs $\sigma \lesssim 0.22\%$. At ordinary published precision (1–3%), the expected outcome is therefore a null score on the stake itself.

Two coarser clauses remain live at ordinary precision: a determination outside $[0.330, 0.370]$ falsifies that program’s dark-energy sector, and the best current inference from measured neighbours is 0.39 ± 0.05 , centred *above* the stake. Whether those clauses matter is the program’s problem. The field’s problem is the missing $N_f = 3$ number.

V. CONCLUSION

No published $T_c/\sqrt{\sigma}$ exists for SU(2) with $N_f = 3$ light fundamental flavours. The calculation is conventional, well posed, and discretization-sensitive. Filling the gap would answer a clean question about flavour dependence in a pseudo-real theory, independent of any particular beyond-Standard-Model motivation. We have stated the requirements, the literature anchors at $N_f = 2$, and the precision a pre-registered stake would need. The computation itself remains to be done.

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